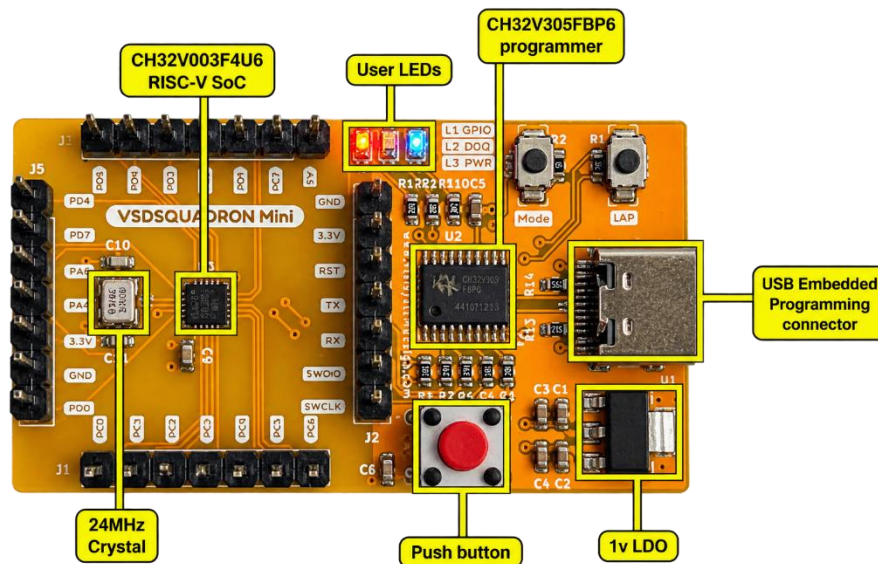


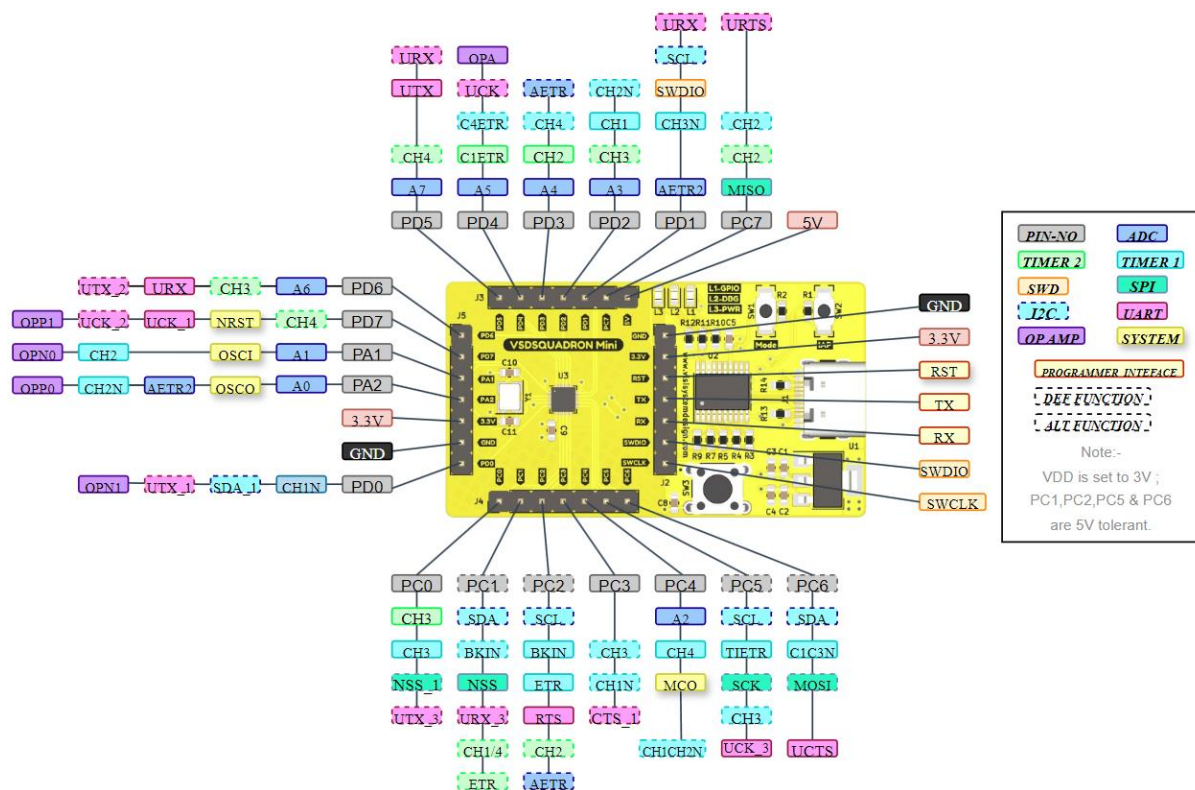
VSDSquadron Mini – Platformio IDE Examples

VSDSquadron Mini

VSDSquadron Mini is a compact RISC-V based development board built around the CH32V003 microcontroller, used for embedded C programming, GPIO control, UART communication, and hardware development.



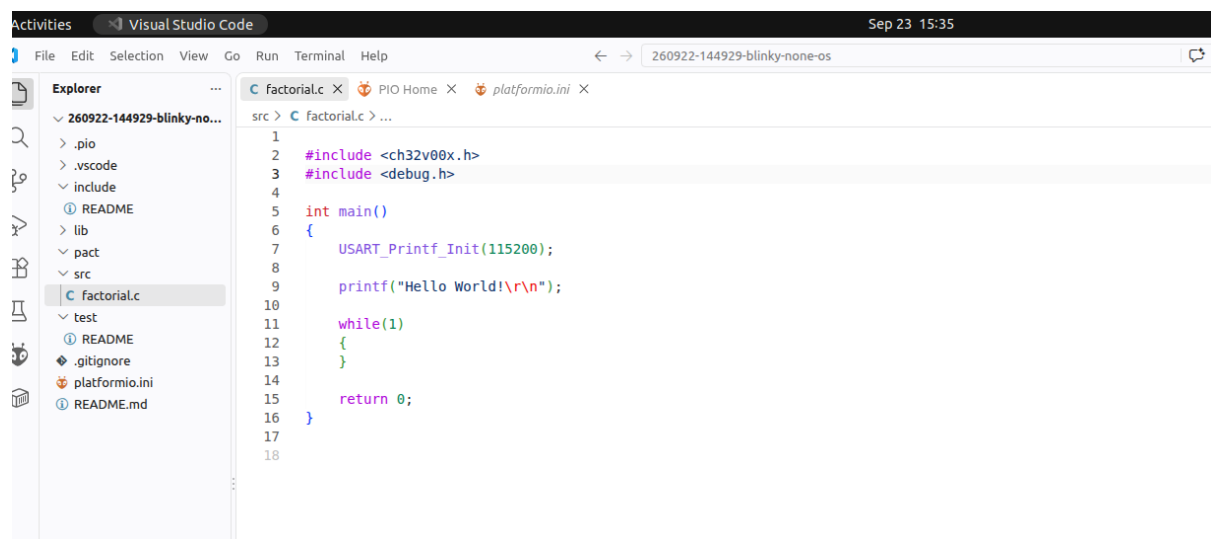
VSDSquadron Mini Pin Diagram



General Workflow

1. Write or edit the .c file inside the project's src/ directory.
2. Compile the project using: `pio run`
3. Upload firmware using: `pio run -t upload`
4. Check the serial interface using: `pio device list`
5. For UART output, open: `pio device monitor -p /dev/ttyACM0 -b 115200`
6. Press RESET on the board when required to restart the program and observe output.

Example 1-HELLO WORLD



Program Execution commands

- 1) Build -> `pio run`

This command compiles and links the C program. If there are no errors, the terminal displays

```
Problems Output Debug Console Terminal Ports
• vsduser@nagarjuna:~/Documents/PlatformIO/Projects/260922-144929-blinky-none-os$ pio run
Processing vdsquadronMini (board: vdsquadronMini; platform: ch32v; framework: noneos-sdk)
-----
Verbose mode can be enabled via `-v, --verbose` option
CONFIGURATION: https://docs.platformio.org/page/boards/ch32v/vdsquadronMini.html
PLATFORM: VDSquadron (1.0.0+sha.ad19fb6) > VDSquadron Mini
HARDWARE: CH32V003F4U6 48MHz, 2KB RAM, 16KB Flash
DEBUG: Current (wch-link) On-board (wch-link) External (minichlink)
PACKAGES:
- framework-wch-noneos-sdk @ 2.30000.0+sha.34b1b78
- tool-openocd-riscv-wch @ 2.1100.260228 (11.0)
- toolchain-riscv @ 1.80200.190731+sha.e8e7ba9
LDF: Library Dependency Finder -> https://bit.ly/configure-pio-ldf
LDF Modes: Finder ~ chain, Compatibility ~ soft
Found 0 compatible libraries
Scanning dependencies...
No dependencies
Building in release mode
Checking size .pio/build/vdsquadronMini/firmware.elf
Advanced Memory Usage is available via "PlatformIO Home > Project Inspect"
RAM: [====] 20.1% (used 412 bytes from 2048 bytes)
Flash: [====] 28.7% (used 4704 bytes from 16384 bytes)
===== [SUCCESS] Took 0.49 seconds =====
```

2) Upload -> pio run -t upload

This command uploads the compiled firmware to the VDSquadron Mini through the WCH-Link programmer. A successful upload display

```
Problems Output Debug Console Terminal Ports
vsduser@nagarjuna:~/Documents/PlatformIO/Projects/260922-14
vsduser@nagarjuna:~/Documents/PlatformIO/Projects/260922-14
vsduser@nagarjuna:~/Documents/PlatformIO/Projects/260922-144929-blinky-none-os$ pio run -t upload
no dependencies
Building in release mode
Checking size .pio/build/vdsquadronMini/firmware.elf
Advanced Memory Usage is available via "PlatformIO Home > Project Inspect"
RAM: [====] 20.1% (used 412 bytes from 2048 bytes)
Flash: [====] 28.7% (used 4704 bytes from 16384 bytes)
Configuring upload protocol...
AVAILABLE: wch-link
CURRENT: upload_protocol = wch-link
Uploading .pio/build/vdsquadronMini/firmware.elf
Open On-Chip Debugger 0.11.0+dev-snapshot (2026-02-28-11:01)
Licensed under GNU GPL v2
For bug reports, read
    http://openocd.org/doc/doxygen/bugs.html
debug_level: 1

Warn : Transport "sdi" was already selected
Ready for Remote Connections
[wch_riscv.cpu.0] Target successfully examined.
** Programming Started **
** Programming Finished **
** Verify Started **
** Verified OK **
** Resetting Target **
shutdown command invoked
===== [SUCCESS] Took 42.06 seconds =====
```

3) Monitor the output -> pio device monitor -p /dev/ttyACM0 -b 115200

This opens the serial monitor at **115200 baud**.

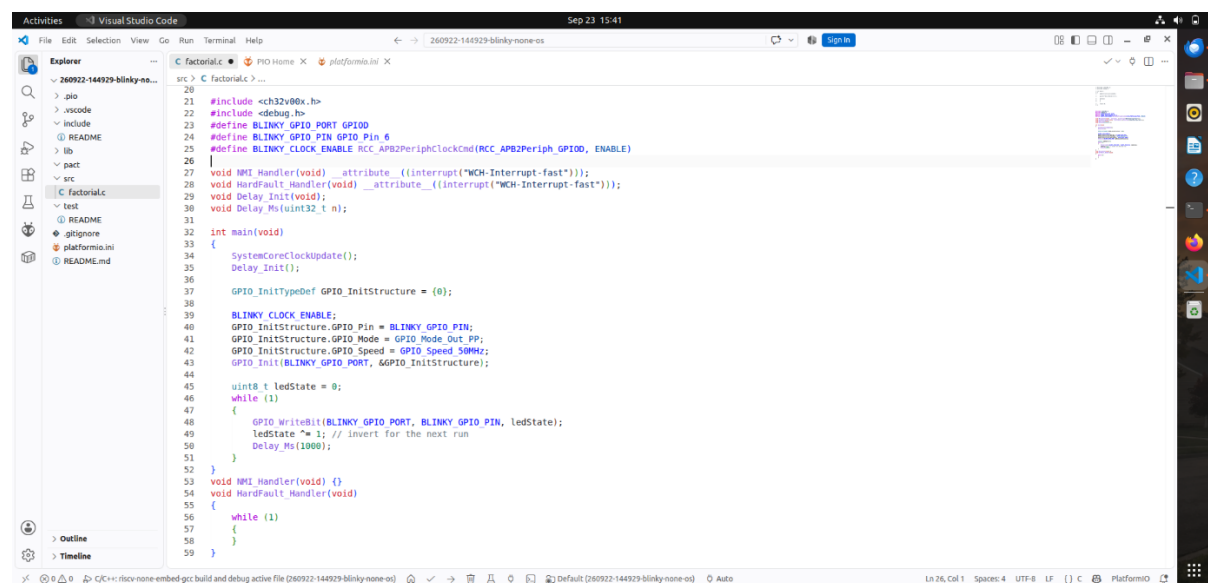
After opening the serial monitor, **press the RESET button on the VSDSquadron Mini once**.
The microcontroller restarts and the program output is displayed in the serial monitor.

```
vsduser@nagarjuna:~/Documents/PlatformIO/Projects/260922-144929-blinky-none-os$ pio device monitor -p /dev/ttyACM0 -b 115200
--- Terminal on /dev/ttyACM0 | 115200 8-N-1
--- Available filters and text transformations: debug, default, direct, hexlify, log2file, nocontrol, printable, send_on_enter, time
--- More details at https://bit.ly/pio-monitor-filters
--- Quit: Ctrl+C | Menu: Ctrl+T | Help: Ctrl+T followed by Ctrl+H
Hello World!
█
```

red-gcc build and debug active file (260922-144929-blinky-none-os) ✓ → 🗑️ 🔧 ⚙️ Default (260922-144929-blinky-none-os) 🔧 Auto

Output:- Hello World!

Example 2- Blink LED



1) Build -> pio run

This command compiles and links the C program. If there are no errors, the terminal displays

```
Problems Output Debug Console Terminal Ports
vdsuser@nagarjuna:~/Documents/PlatformIO/Projects/260922-144929-blinky-none-os$ pio run
Processing vdsquadronMini (board: vdsquadronMini; platform: ch32v; framework: noneos-sdk)
-----
Verbose mode can be enabled via '-v, --verbose' option
CONFIGURATION: https://docs.platformio.org/page/boards/ch32v/vdsquadronMini.html
PLATFORM: VDSquadron (1.0.0+sha.ad19fb6) > VDSquadron Mini
HARDWARE: CH32V003F4U6 48MHz, 2KB RAM, 16KB Flash
DEBUG: Current (wch-link) On-board (wch-link) External (minichlink)
PACKAGES:
- framework-wch-noneos-sdk @ 2.30000.0+sha.34b1b78
- tool-openocd-riscv-wch @ 2.1100.260228 (11.0)
- toolchain-riscv @ 1.80200.190731+sha.e8e7ba9
LDF: Library Dependency Finder -> https://bit.ly/configure-pio-ldf
LDF Modes: Finder ~ chain, Compatibility ~ soft
Found 0 compatible libraries
Scanning dependencies...
No dependencies
Building in release mode
Compiling .pio/build/vdsquadronMini/src/factorial.o
Linking .pio/build/vdsquadronMini/firmware.elf
Checking size .pio/build/vdsquadronMini/firmware.elf
Advanced Memory Usage is available via "PlatformIO Home > Project Inspect"
RAM: [=====] 13.9% (used 284 bytes from 2048 bytes)
Flash: [=====] 7.7% (used 1264 bytes from 16384 bytes)
Building .pio/build/vdsquadronMini/firmware.bin
===== [SUCCESS] Took 0.68 seconds =====
```

2) Upload -> pio run -t upload

This command uploads the compiled firmware to the VDSquadron Mini through the WCH-Link programmer. A successful upload display

```
Problems Output Debug Console Terminal Ports
vdsuser@nagarjuna:~/Documents/PlatformIO/Projects/260922-144929-blinky-none-os$ pio run -t upload
-----
Scanning dependencies...
No dependencies
Building in release mode
Checking size .pio/build/vdsquadronMini/firmware.elf
Advanced Memory Usage is available via "PlatformIO Home > Project Inspect"
RAM: [=====] 13.9% (used 284 bytes from 2048 bytes)
Flash: [=====] 7.7% (used 1264 bytes from 16384 bytes)
Configuring upload protocol...
AVAILABLE: wch-link
CURRENT: upload protocol = wch-link
Uploading .pio/build/vdsquadronMini/firmware.elf
Open On-Chip Debugger 0.11.0+dev-snapshot (2026-02-28-11:01)
Licensed under GNU GPL v2
For bug reports, read
http://openocd.org/doc/doxygen/bugs.html
debug_level: 1

Warn : Transport "sdi" was already selected
Ready for Remote Connections
[wch_riscv.cpu.0] Target successfully examined.
** Programming Started **
** Programming Finished **
** Verify Started **
** Verified OK **
** Resetting Target **
shutdown command invoked
===== [SUCCESS] Took 23.76 seconds =====
vdsuser@nagarjuna:~/Documents/PlatformIO/Projects/260922-144929-blinky-none-os$
```

The LED Blink program was compiled using pio run and uploaded using pio run -t upload; after successful programming, the LED was observed blinking on the VDSquadron Mini.

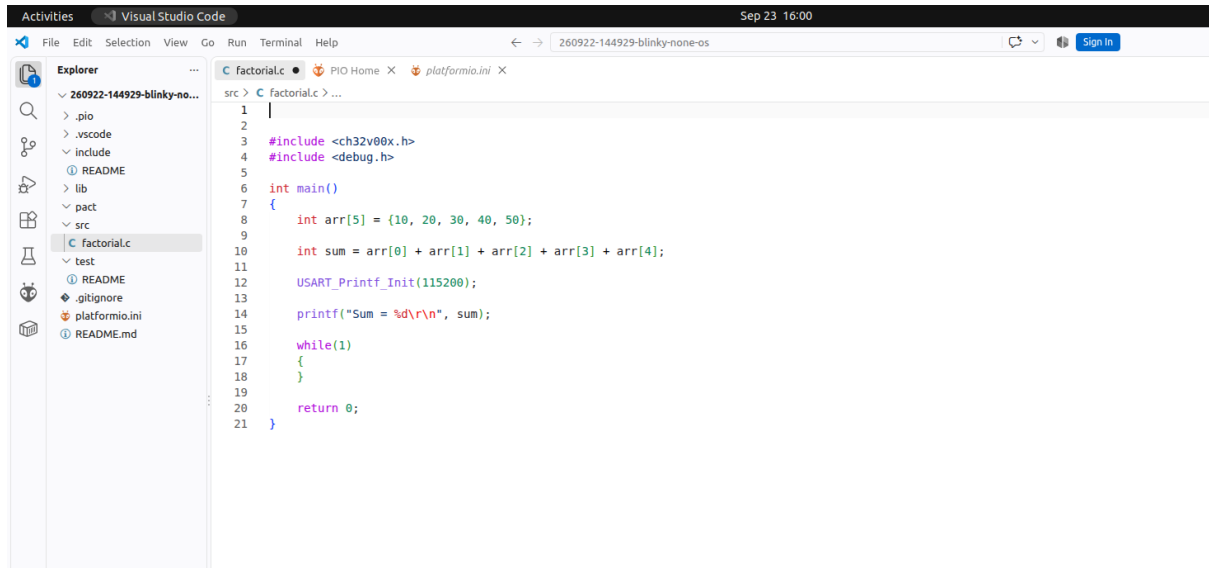


Before uploading the code



after uploading the code

Example 3 – Array Sum



The screenshot shows the Visual Studio Code interface. The Explorer panel on the left shows a project named '260922-144929-blinky-none-os' with a file named 'Factorial.c'. The main editor displays the contents of 'Factorial.c', which is a C program that calculates the sum of an array. The code is as follows:

```
1 |
2 |
3 | #include <ch32v00x.h>
4 | #include <debug.h>
5 |
6 | int main()
7 | {
8 |     int arr[5] = {10, 20, 30, 40, 50};
9 |
10 |    int sum = arr[0] + arr[1] + arr[2] + arr[3] + arr[4];
11 |
12 |    USART_Printf_Init(115200);
13 |
14 |    printf("Sum = %d\r\n", sum);
15 |
16 |    while(1)
17 |    {
18 |    }
19 |
20 |    return 0;
21 | }
```

1) Build -> pio run

This command compiles and links the C program. If there are no errors, the terminal displays



The screenshot shows the terminal output of the 'pio run' command. The output is as follows:

```
vsduser@nagarjuna:~/Documents/PlatformIO/Projects/260922-144929-blinky-none-os$ pio run
Processing vdsquadronMini (board: vdsquadronMini; platform: ch32v; framework: noneos-sdk)
-----
Verbose mode can be enabled via '-v', '--verbose' option
CONFIGURATION: https://docs.platformio.org/page/boards/ch32v/vdsquadronMini.html
PLATFORM: VDSquadron (1.0.0+sha.ad19fb6) > VDSquadron Mini
HARDWARE: CH32V003F4U6 48MHz, 2KB RAM, 16KB Flash
DEBUG: Current (wch-link) On-board (wch-link) External (minichlink)
PACKAGES:
- framework-wch-noneos-sdk @ 2.30000.0+sha.34b1b78
- tool-openocd-riscv-wch @ 2.1100.260228 (11.0)
- toolchain-riscv @ 1.80200.190731+sha.e8e7ba9
LDF: Library Dependency Finder -> https://bit.ly/configure-pio-ldf
LDF Modes: Finder ~ chain, Compatibility ~ soft
Found 0 compatible libraries
Scanning dependencies...
No dependencies
Building in release mode
Checking size .pio/build/vdsquadronMini/firmware.elf
Advanced Memory Usage is available via "PlatformIO Home > Project Inspect"
RAM:   [=          ] 13.9% (used 284 bytes from 2048 bytes)
Flash: [=          ]  7.7% (used 1264 bytes from 16384 bytes)
===== [SUCCESS] Took 0.51 seconds =====
```

2) Upload -> pio run -t upload

This command uploads the compiled firmware to the VSDSquadron Mini through the WCH-Link programmer. A successful upload display

```
vsduser@nagarjuna:~/Documents/PlatformIO/Projects/260922-144929-blinky-none-os$ pio run -t upload
Scanning dependencies...
No dependencies
Building in release mode
Checking size .pio/build/vsdsquadronMini/firmware.elf
Advanced Memory Usage is available via "PlatformIO Home > Project Inspect"
RAM: [=====] 13.9% (used 284 bytes from 2048 bytes)
Flash: [=====] 7.7% (used 1264 bytes from 16384 bytes)
Configuring upload protocol...
AVAILABLE: wch-link
CURRENT: upload_protocol = wch-link
Uploading .pio/build/vsdsquadronMini/firmware.elf
Open On-Chip Debugger 0.11.0+dev-snapshot (2026-02-28-11:01)
Licensed under GNU GPL v2
For bug reports, read
  http://openocd.org/doc/doxygen/bugs.html
debug_level: 1

Warn : Transport "sdi" was already selected
Ready for Remote Connections
[wch_riscv.cpu.0] Target successfully examined.
** Programming Started **
** Programming Finished **
** Verify Started **
** Verified OK **
** Resetting Target **
shutdown command invoked

===== [SUCCESS] Took 23.34 seconds =====
```

3) Monitor the output -> pio device monitor -p /dev/ttyACM0 -b 115200

This opens the serial monitor at **115200 baud**.

After opening the serial monitor, **press the RESET button on the VSDSquadron Mini once**. The microcontroller restarts and the program output is displayed in the serial monitor.

```
vsduser@nagarjuna:~/Documents/PlatformIO/Projects/260922-144929-blinky-none-os$ pio device monitor -p /dev/ttyACM0 -b 115200
--- Terminal on /dev/ttyACM0 | 115200 8-N-1
--- Available filters and text transformations: debug, default, direct, hexlify, log2file, nocontrol, printable, send_on_enter, time
--- More details at https://bit.ly/pio-monitor-filters
--- Quit: Ctrl+C | Menu: Ctrl+T | Help: Ctrl+T followed by Ctrl+H
Sum = 150
█

14929-blinky-none-os  ✓ → 🗑️ 📄 🔍 🔄 Default (260922-144929-blinky-none-os) 🔄 Auto Ln 1, Co
```

Output :- Sum = 150